

Semester-wise Pattern for Students Admitted to M.Sc. Semiconductor and Nanoscience Programme								
Year	First Semester			U	Second Semester			U
I	BITS	F103	Introduction to Engineering Design and Prototyping	4	BITS	F111	Thermodynamics	3
	BIO	F101	Introduction to Biological Sciences	3	BITS	F112	Technical Report Writing	2
	CHEM	F101	Fundamentals of Chemistry	3	CS	F111	Computer Programming	4
	MATH	F101	Multivariable Calculus	3	MATH	F113	Probability and Statistics	3
	PHY	F101	Oscillations and Waves	3	MATH	F102	Linear Algebra and Complex Variables	3
	BITS	F101	Social Conduct	1	EEE	F111	Electrical Sciences	3
	BITS	K101	Physical Well-being and Creativity	1*	BITS	F102	Innovation and Design Thinking	1
				17				19
II	MATH	F211	Mathematics III	3	SNS	F241	Chemistry at Nanoscale	3
	SNS	F211	Fundamentals of Nanoscience and Nanotechnology	3	SNS	F242	Electronic Devices	3
	SNS	F212	Quantum Mechanics	3	SNS	F243	Materials & Nanoscale Characterization	4
	SNS	F213	Material Science and Engineering	3	SNS	F244	Nanomaterials Synthesis & Characterization Lab	3
	SNS	F214	Fundamentals of Solid State	3	MGTS	F211	Principles of Management	3
	BITS	F225	Environmental Studies	3	ECON	F211	OR	5
			Humanities Elective	3			Principles of Economics Humanities Elective	
				21				21
Summer BITS F221 Practice School – I								
III			Open/Humanities Electives	3 to 6			Open Electives	3
	SNS	F311	Semiconductor Thin Film Technology	3	SNS	F341	Principles of Nanobiotechnology	3
	SNS	F312	Molecular Simulations	3	SNS	F342	Principles of Nanophotonics and Devices	3
	SNS	F313	Instrumental Methods of Analysis	4	SNS	F343	Advanced Nanofabrication	3
			Discipline Electives	6			Discipline Elective	9
				19/22				21
IV	Open Electives			8 to 14	BITS	F412	Practice School-II	20
						or	or	
					BITS	F421T	Thesis	16
				or	F425T	Thesis	18-20	
				or	F424T	Thesis (9) and Electives	15-18	
					(6 to 9)			
				8/14				15/20

Discipline core – **44** Units (**14** Courses)

Discipline elective – **15** Units (**4** Courses (min))